

Assessment of I-4 Contraflow Plans

Microscopic Versus Mesoscopic Simulation

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Better use of the available road network is critical to improving the evacuation operation during a disaster. Contraflow operations help increase the capacity of the available network by reversing the direction of inbound lanes to outbound lanes. This helps improve the outflow from a region threatened by disaster. One of the major issues associated with contraflow operations is determining the locations for access to the contraflow lanes from the normal-flow lanes. These accesses are also referred to as crossovers. Four different strategies with different crossover locations were tested on the I-4 evacuation route from Tampa to Orlando, Florida. It was found that the provision of two crossovers, one after Tampa and another after Plant City, performed the best but was only marginally better than the provision of one crossover after Tampa. Therefore, considering the cost and personnel needed to provide a crossover, the provision of one crossover after Tampa was found to be a more logical choice than the provision of two crossovers. It was observed that the time required to run the microscopic simulation to arrive at the results was extremely long. To overcome this drawback, the cell transmission model (CTM) was calibrated and run for the same four strategies. It was observed that the results were extremely close to the results from the microscopic simulation. The robustness and speed of CTM make it ideal for use as part of a decision support system to help determine the best strategies in real time. This will help emergency management officials make real-time decisions in the event of unforeseen drops in capacities because of incidents or vehicle breakdowns.

Contraflow operations are widely being used to improve evacuation operations. Contraflow operation is the reversal of inbound lanes to outbound lanes to increase the capacities of evacuation routes. One of the main issues related to contraflow operations is access to contraflow lanes. These accesses are also referred to as crossovers. Placement of these crossovers at inappropriate locations might result in bottlenecks, which adversely affect evacuation operations. Therefore, it is of utmost importance to determine locations for crossovers that will help improve evacuation operations. The number of crossovers is also constrained by the amount of personnel required and the logistics and cost required to implement them. Hence, it is crucial to determine the locations and number of crossovers under constraints that will provide the best benefit to the evacuation operation.

Work done by Wolshon helped provide an understanding of contraflow operations during an evacuation (1). That work discussed the

safety, operational, accessibility, and cost issues associated with contraflow lanes. That work also tabulated the gains in capacity for each lane from the conversion of lanes to contraflow operations. The paper pointed out that when all the lanes were reversed, the total increase in capacity was only 67%, although a 100% increase in capacity should have been achieved because of the reversal of all lanes. This less than 100% increase in capacity was attributed to the unfamiliarity and discomfort of drivers driving on the contraflow lane, resulting in lower speeds.

Wolshon and Urbina published a comprehensive review of the evacuation plans of all states affected by hurricanes (2). They recommended the need to control evacuation demand and increase the capacity of evacuation routes and develop a standard set of evacuation guidelines. They suggested that the evacuation plans need to be more flexible to handle eventualities.

Tuydes and Ziliaskopoulos proposed the use of an optimization model to determine sections of roads where contraflow plans need to be implemented (3). One of the drawbacks was that the results generated with this model did not incorporate merge and diverge congestions, which might be crucial in determining the location of crossovers.

Theodoulou (4) and Theodoulou and Wolshon (5) modeled and analyzed certain freeway contraflow operations in the New Orleans, Louisiana, area. Theirs was one of the first efforts to discuss design issues with respect to entry–exit and termination strategies. The basic idea that they used was to manage the congestion caused by merge and diverge and improve the use of the capacity created because of contraflow operations. In his study, Theodoulou had assumed 15% heavy vehicles (4). As stated in his research (4), he assumed a high percentage of heavy vehicles, because it has been observed in previous studies (6, 7) that evacuees instinctively tend to carry as many of their belongings as possible. Theodoulou also observed that 60% of evacuees tend to use normal-flow lanes, whereas the remaining 40% use the contraflow lanes (4).

The road signs and markings on the contraflow lanes are created for vehicles traveling in the opposite direction of the evacuating traffic. This results in driver unfamiliarity with the driving conditions, resulting in a drop in free-flow speeds. For the purpose of this study, it was assumed that the free-flow speeds on the contraflow lanes dropped from 60 to 55 mph. Discomfort is also associated with driving on the contraflow lanes because of the limited number of exits on the contraflow lanes, which provide limited opportunities to stop for rest. The unfamiliarity and discomfort associated with traveling on contraflow lanes would result in fewer evacuees preferring to use contraflow lanes than normal-flow lanes. The rates of use of the contraflow lanes compared with the rates of use of the normal-flow lanes were assumed to be similar to the values observed by Theodoulou (4): 60% evacuees use normal-flow lanes, whereas the remaining 40% use the contraflow lanes.

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TABLE 1 Demand from Each Crossover Location to Orlando on I-4

City	Total Vehicles	Entry Flow Rate (veh/h)	Time Period (h)
Tampa	67,000	3,600	18.61
Plant City	12,000	1,000	12
Lakeland	33,000	2,400	13.75

The present study evaluated the contraflow plans for Interstate-4, which is a major evacuation route from Tampa to Orlando, Florida. A Post, Buckley, Schuh, & Jernigan, Inc., report on contraflow operations for Interstate-4 stated that the contraflow operation is to operate from Tampa to the Orange County, Florida, line (8). This was used as the guideline for the contraflow operations on I-4. Under free-flow conditions, it takes 1.5 h to travel from Tampa (I-4 Milepost 1) to the start of the Orange County line in Orlando.

Plant City and Lakeland were identified as the other two major demand origins along I-4. I-4 has three lanes in each direction, with a free-flow speed of 60 mph. All westbound lanes were reversed during contraflow operations. As discussed earlier, the free-flow speed on the contraflow lanes was assumed to be 55 mph. The demands for the region were estimated on the basis of the number of households from U.S. census data (9). It was assumed that each household would evacuate with only one vehicle. It was also assumed that 50% of the evacuees from Tampa and 100% from Plant City and Lakeland would evacuate toward Orlando. The demand from each region is shown in Table 1. Vehicles were loaded from the three locations on the basis of fixed entry flow rate for a predetermined time. Time period is defined as the demand input period for all the vehicles to be evacuated. The entry flow rates and the time for the simulation model are also shown in Table 1.

The analysis for determination of the locations of crossovers was conducted by the use of two distinct simulation approaches. The

first analysis was done with microscopic simulation software called VISSIM (10). The second analysis was done by using the cell transmission model (CTM) proposed by Daganzo (11, 12). The results from both analyses are compared to determine the appropriate locations for the crossover point.

MICROSCOPIC SIMULATION ANALYSIS

The traffic network (Figure 1) was coded in VISSIM software as links and connectors by using aerial photographs as overlays. Three possible crossover locations were available for this study area. The first feasible crossover location was 4 mi from the start of I-4, the second crossover location (Plant City) was 46 mi from the first feasible crossover location, and the last feasible crossover location (Lakeland) was 16 mi from the crossover at Plant City. Under free-flow conditions, it takes 30 min to travel from the crossover at Tampa to the crossover at Plant City and it takes 15 min to travel from the crossover at Plant City to the crossover at Lakeland.

To determine the best crossover locations, four strategies were compared with each other and the base case, which is normal operations with no contraflow lanes. The four strategies are as follows:

Strategy 1. One crossover after Tampa,

Strategy 2. Two crossovers, one after Tampa and the other after Plant City,

Strategy 3. Two crossovers, one after Plant City and the other after Lakeland, and

Strategy 4. Two crossovers, one after Tampa and the other after Lakeland.

The crossover locations are shown in Figure 1. For all of these strategies with contraflow operations, the contraflow lanes were terminated by making the normal-flow lanes exit and then transferring vehicles from the contraflow lanes to the normal-flow lanes. Also, all the crossovers discussed in the strategies provide access for vehicles on the normal-flow lanes so that they may use the contraflow lanes. This was done to ensure that there was no

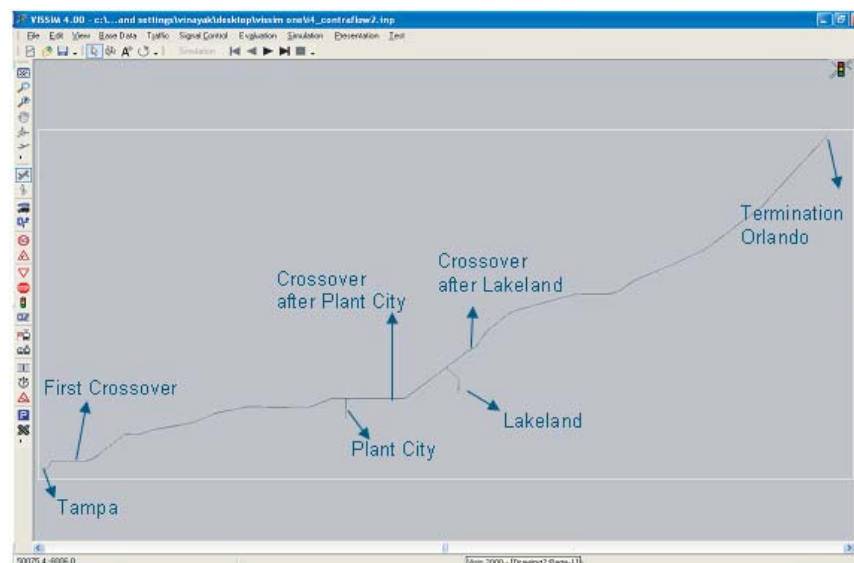


FIGURE 1 VISSIM-coded Map of I-4 from Tampa to Orlando.

TABLE 2 VISSIM Simulation Results for Base Case and Four Evacuation Clearance Time Strategies

Microscopic Simulation	Total Travel Time (h)	Evacuation Clearance Time (h)				Excess Vehicles at Lakeland
		<i>t</i> 80	<i>t</i> 85	<i>t</i> 90	<i>t</i> 95	
Base case	134,012.9	14.77	15.65	16.99	18.39	13,014
Strategy 1	113,179.7	13.76	14.91	16.53	18.16	0
Strategy 2	112,056.1	13.73	14.86	16.53	18.19	0
Strategy 3	119,408.3	14.18	15.28	16.78	18.26	8,840
Strategy 4	114,301	14.1	15.22	16.74	18.27	11,135

NOTE: *t*80, *t*85, *t*90, and *t*95 = 80%, 85%, 90%, and 95% evacuation clearance times, respectively.

congestion because of the merging of vehicles at the termination of the contraflow lanes. Merging at the termination of contraflow could result in huge backups and reduce the benefits of contraflow operations.

Vehicles were loaded from the three locations on the basis of a fixed entry flow rate for a predetermined time. The entry flow rates and times for the simulation model are shown in Table 1. Because of congestion at the Lakeland on-ramp, the simulation software was unable to load the vehicles on the on-ramp according to the pre-specified flow during the prespecified time. This resulted in the inability of some vehicles to enter the network. Therefore, the number of vehicles unable to enter the network is used as a measure of performance of congestion.

Multiple measures of effectiveness (MOEs) were considered for comparison of the base case and the four strategies. The MOEs used were the total travel time of all the vehicles; the number of vehicles unable to enter the network; and the 80%, 85%, 90%, and 95% evacuation clearance times. Loop detectors were placed at the destination link to count the 80%, 85%, 90%, and 95% of vehicles exiting the network. The VISSIM microscopic simulation results for all the scenarios are shown in Table 2.

As can be seen in Figure 2a, Strategy 1 and Strategy 2 performed the best, with all vehicles from all three regions being loaded onto the network. The base case performed the worst, with an average of approximately 13,000 vehicles being unable to enter the network at Lakeland. Approximately 11,134 and 8,840 vehicles were unable to enter the network for Strategies 4 and 3, respectively. It should be noted that the availability of one crossover (Strategy 1) performed better than the availability of two crossovers (Strategies 3 and 4).

It was observed that more vehicles were unable to enter the network at Lakeland in Strategy 4 than in Strategy 3. This is because the crossover alleviates the merge congestion where on-ramps merge with normal-flow lanes. In Strategy 3, the upstream crossover is located at Plant City, whereas in Strategy 4 the upstream crossover is located at Tampa. In Strategy 4, the vehicles entering at Plant City do not get an opportunity to access the contraflow lanes, which adds to the flow on the normal-flow lanes compared with the flow on those lanes in Strategy 3, in which vehicles get distributed to the contraflow lanes. Therefore, because of the higher flows on the normal-flow lanes in Strategy 4, the congestion at the merge at Lakeland is higher, resulting in a larger number of vehicles not being able to enter the network at Lakeland.

Another interesting aspect observed was the shockwave effect because of the slower speeds and the lower flow rates at the crossover.

This effect occurred at the crossover after Lakeland, where a large number of vehicles could not enter the network at Lakeland in Strategy 3, whereas in Strategy 1, all the vehicles were able to enter the network at Lakeland. This difference cannot be attributed to the merge congestion, because the flows on the normal-flow lanes after Plant City in Strategy 1 and Strategy 3 were the same, but can be attributed only to the shockwave effects because of the downstream crossover at Lakeland. The downstream crossover after Plant City does not result in excess vehicles left at Plant City because of the low flows entering at Plant City.

Total vehicle travel time is an important MOE for an evacuation plan. A lower total vehicle travel time indicates a reduction of the amount of time that vehicles spend in the network. Figure 2b shows the total vehicle travel time for the base case and the four strategies. Even though all the demand was not loaded onto the network, Strategy 2 had the least total vehicle travel time and performed only marginally better than Strategy 1.

Evacuation clearance time is an important MOE for evacuation strategies. It is a measure of how quickly people are able to get out of the threat zone. Four different measures of evacuation clearance time were considered: the times required to clear 80%, 85%, 90%, and 95% of the evacuees who reached the destination. From Figure 2c, it can be concluded that Strategy 2 performed the best but was only marginally better than Strategy 1. Figure 2c also shows that as the proportion of evacuees reaching the destination increases from 80% to 95%, the difference between the clearance times between the strategies decreases. This is observed as the flattening of the dip in Strategy 1 and Strategy 2 as the percentage of evacuees reaching the destination increases. This is because as the percentage of evacuees reaching their destination increases, there are origins that were totally evacuated and no vehicles that could affect operations are entering from those origins. Hence, the strategies that helped reduce the congestion were not being effective, because there was no congestion to be alleviated in the later stages (when the percentage of evacuees reaching their destination increased). For example, after 13.75 h all vehicles from both Plant City and Lakeland had left their respective origins, after which there were no merge congestions at these two on-ramps. A total of about 86% of the vehicles had left their respective origins after 13.75 h, after which there would be no congestion at these on-ramps; therefore, it is no coincidence that a significant flattening of the dip was observed from 85% to 90% evacuee clearance. The minute difference for the evacuation clearance times for 90% and 95% of the evacuees reaching their destination still observed in the strategies was due to the shockwave effect.

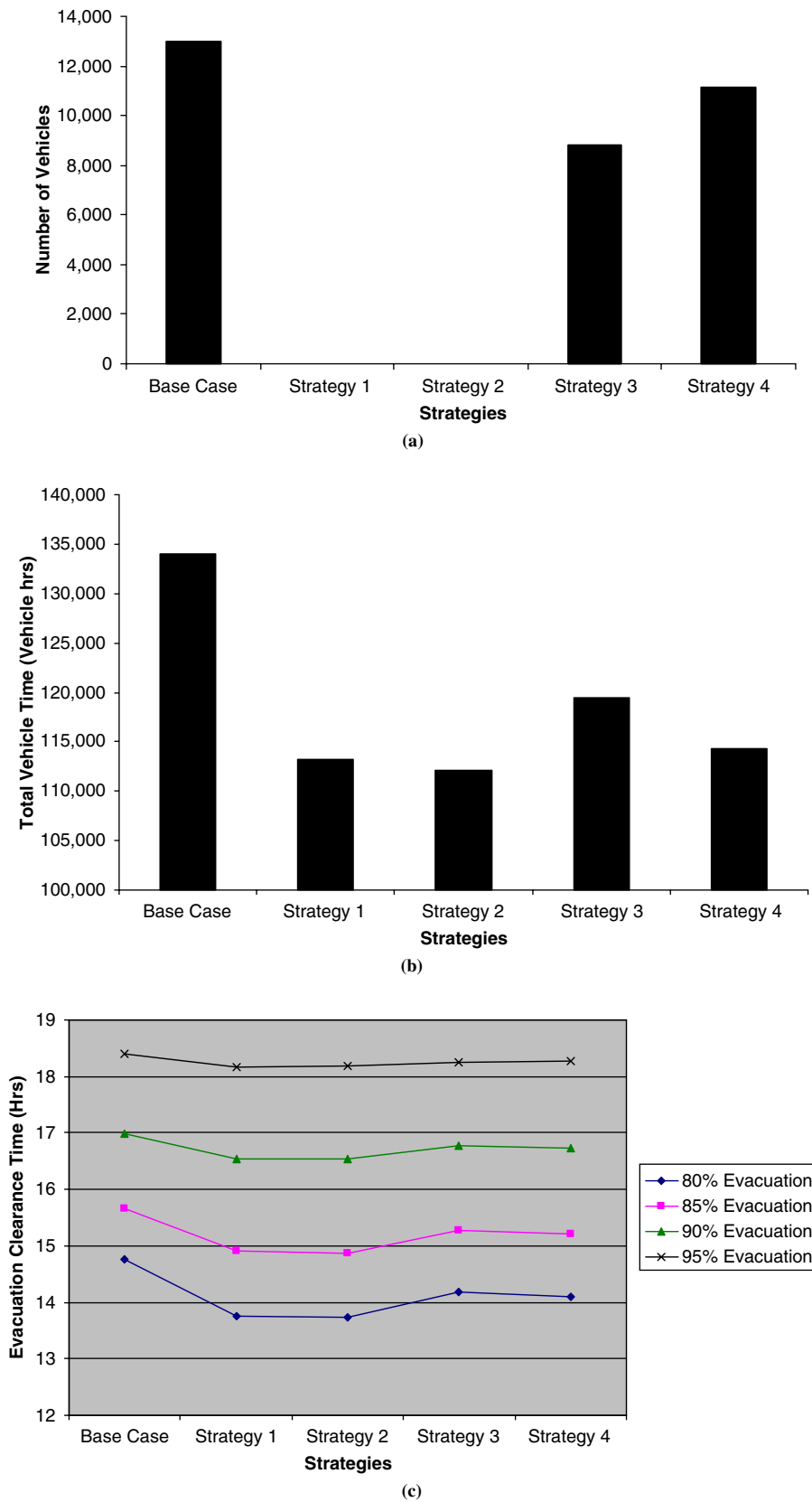


FIGURE 2 MOEs for various evacuation clearance time strategies obtained by using VISSIM microscopic simulation: (a) number of vehicles not dissipated into the network for various strategies, (b) total vehicle travel time for various strategies, and (c) 80%, 85%, 90%, and 95% evacuation clearance times for various strategies.

The results show that the provision of one crossover after Tampa and another after Plant City provided the best evacuation operations but was only marginally better than the provision of one crossover at Tampa. It was also observed that the provision of one crossover after Tampa was better than the provision of two crossovers, corresponding to Strategy 4 and Strategy 5, because of merge and diverge congestion.

It was observed that microscopic simulation analysis for such a simple network structure took an extremely large amount of time (4 h) for a single run on a Pentium 4 processor with 2 GB of RAM on a Windows XP machine. This was a huge drawback to the use of this approach. In the case of bottlenecks created by unforeseen events leading to a drop in capacity (such as incidents and breakdown of vehicles), it is crucial to make real-time decisions to ensure the smooth flow of traffic. These decisions might include having to create a new location for a crossover. Because approximately 4 h of time is taken to execute one run for a scenario, at least five runs are required to come to some kind of statistical confidence, and because five scenarios need to be tested, simple number crunching leads to the conclusion that approximately 100 computer hours is required to decide on a strategy. Real-time decisions cannot be made with such a long computing time. This problem created a need to explore other simulation models. CTM, a mesoscopic model, seemed to hold the solution to this problem. The nice linear relationships between flows and occupancies made it seem to be an ideal candidate for use for real-time simulation. The deterministic nature of this simulation also resulted in doing away with the need for multiple runs to evaluate a strategy.

CELL TRANSMISSION MODEL

The CTM proposed by Daganzo is an interesting way to model traffic flow in a network and in a freeway corridor (11, 12). CTM is a deterministic mesoscopic model that predicts flow, occupancy, and density with accuracy. Unlike macroscopic models, CTM is able to account for backward wave propagation because of congestions downstream and is also able to predict the merge and diverge behaviors of traffic at junctions. Unlike microscopic simulations, CTM consumes far less time and space on the computer and is much easier to code. This makes CTM an ideal tool for use as an underlying basis for a decision support system that will help decision makers determine quickly and accurately the best possible strategies to be used during an evacuation.

In CTM, the road section is divided into homogeneous cells, such that the length of each cell is the time that it takes for a vehicle at free-flow speed to traverse the length of that section in one clock tick (time unit). The state of the system at any time t is determined by the number of vehicles in cell i and is expressed as $n_i(t)$. Parameter $N_i(t)$ defines the maximum number of vehicles allowed in cell i during time interval t . Parameter $Q_i(t)$ defines the maximum number of vehicles that can flow into cell i when the clock advances from t to $t + 1$. If the cells are numbered consecutively starting with the upstream end of the road from $i = 1$ to I , the recursive relationship of CTM can be expressed as

$$n_i(t+1) = n_i(t) + y_i(t) - y_{i+1}(t) \quad (1)$$

where $y_i(t)$ is the inflow to cell i in the time interval $(t, t + 1)$, given by

$$y_i(t) = \min\{n_{i-1}(t), Q_i(t), \delta[N_i(t) - n_i(t)]\} \quad (2)$$

where δ is equal to w/v , where w is the speed with which disturbances propagate backward when traffic is congested and v is the free-flow speed. It was seen by Daganzo (11, 12) that the accuracy of the model is enhanced if δ is redefined such that

$$\delta = \begin{cases} 1 & \text{if } n_{i-1}(t) \leq Q_i(t) \\ \frac{w}{v} & \text{if } n_{i-1}(t) > Q_i(t) \end{cases} \quad (3)$$

CTM can be formulated as a linear programming problem (3, 13) that can provide guidelines for traffic assignment and signal timings.

CALIBRATION OF CTM FOR INTERSTATE 4

The first step for the calibration of CTM for Interstate 4 was to determine the size of the smallest tick of the clock in CTM. To maintain an acceptable amount of resolution, each tick was chosen to be 30 s. Because the free-flow speed on the normal-flow lanes was 60 mph and that on the contraflow lanes was 55 mph, the cell length was determined by multiplying the number of seconds in a tick by the free-flow speed of the type of lanes. Therefore, the cell length of the normal-flow lanes is 0.5 mi, whereas that of the contraflow lanes is 0.46 mi.

Because it was assumed that the proportion of heavy vehicles was 15% during the simulation, the average length of vehicles would be greater than the length of a normal vehicle. The length of a normal vehicle is assumed to be 22 ft, whereas that of a heavy vehicle is assumed to be 40 ft. To determine the average length of vehicles, a weighted average of the lengths of the two types of vehicles was taken on the basis of the percentage of each kind of vehicle on the road. This implies that $0.15 * 40 + 0.85 * 22 = 24.7$ ft is the average length of the vehicles.

The average length of the vehicles, the length of a cell, and the number of lanes in a cell were considered to determine the maximum number of vehicles that a cell can hold. It was found that a cell on normal-flow lanes could hold a maximum of 320 vehicles and a cell on the contraflow lanes could hold a maximum of 293 vehicles.

The maximum flows in each cell were determined by the number of lanes in each cell. Considering that the maximum flow per lane is 1,800 vehicles per hour, it was calculated that 15 vehicles per tick per lane was the maximum flow through a lane in a cell. The cells at the on-ramp from Plant City had four lanes and those at Lakeland had five lanes, in agreement with the network designed in the VISSIM simulation.

For the crossovers, the free-flow speeds were lower than those for the normal-flow lanes and the contraflow lanes. Therefore, the lengths of the cells for crossovers were smaller than those for the other cells.

The parameter δ is used to enhance the modeling realism on free-ways and was assumed to be 1. The reason for this was inspired by the work done by Kalafatas and Peeta (14), in that it does not require the incorporation of the value of δ , since for freeway cells $Q_i(t)/[N_i(t) - Q_i(t)]$ is approximately equal to $\frac{1}{2}$, which is close to the accepted value, which ranges from $\frac{1}{4}$ to $\frac{1}{2}$. For this modeling, $Q_i(t)/[N_i(t) - Q_i(t)]$ is $\frac{1}{2}$ for both the normal and the contraflow lanes.

To mimic the same conditions used in the VISSIM network, excess vehicles at Lakeland were not loaded onto the network after a predetermined time period. The number of vehicles left in

the source cell at Lakeland was recorded but was not loaded onto the network.

RESULTS

Results of CTM for Interstate 4

The results of the simulation executed with CTM are shown in Table 3.

Graphs are plotted for the various strategies and MOEs as the number of vehicles that were unable to be loaded onto the network (Figure 3a); total vehicle travel time (Figure 3b); and the evacuation clearance times for 80%, 85%, 90%, and 95% of the evacuees (Figure 3c). The results from CTM are similar to the results observed in the VISSIM simulation. Strategy 2 was found to have the least total vehicle travel time, but it was only marginally better than that for Strategy 1. It was also found that the evacuation clearance times for 80%, 85%, 90%, and 95% of the evacuees were the same for Strategy 1 and Strategy 2. This was because the difference between the clearance times for Strategy 1 and Strategy 2 from the VISSIM simulation results was less than 1%. Therefore, CTM could not predict the slight difference in clearance times.

Comparison of CTM with VISSIM

A comparison of CTM with VISSIM shows that the errors in estimation of the MOEs (total travel time; evacuation clearance times for 80%, 85%, 90%, and 95% of the evacuees; and the excess number of vehicles that were not able to enter the network from Lakeland) were much less than or close to 5% (Table 4). This indicates that the results obtained with CTM are comparable to those obtained with VISSIM.

It was observed that the computing time for the five scenarios in CTM was 7.5 min, whereas that in VISSIM was 100 h. This indicates that CTM is an ideal tool for use as a real-time decision support system to determine dynamic crossovers because of an unforeseen incident.

It was also observed that VISSIM did not load the vehicles if they could not enter the network during a given time frame. This drawback can easily be handled in CTM, so that more reliable estimates of the evacuation clearance time and total vehicle travel time can be made. The results for all the vehicles loaded onto the network are shown in Table 5. It is seen that 2.6 h is saved in clearing 80% of the evacuees by using Strategy 1. It was also observed that the total vehicle travel time for Strategy 1 was significantly reduced by

16% from that in the base case. The plots of the MOEs for the various strategies by the use of CTM are shown in Figure 4.

CONCLUSIONS

The paper compares two levels of simulation, mesoscopic and microscopic, represented by CTM and VISSIM, respectively. The two simulations were also used to determine the best crossover locations for the evacuation routes on Interstate-4 between Tampa and Orlando.

It was interesting to find that the provision of one crossover after Tampa and another after Plant City provided the best evacuation operations but was only marginally better than the provision of one crossover at Tampa. Therefore, considering the cost and personnel needed to provide a crossover, the provision of one crossover after Tampa was found to be a more logical choice than the provision of two crossovers. It was also observed that the provision of two crossovers, with one of those crossovers being after Lakeland, performed worse than the provision of just one crossover. This shows that the idea that more crossovers is better is wrong and that the location and the number of crossovers needed should be considered.

From the results observed in this study, two main aspects need to be addressed for contraflow planning for a general corridor. The first is that contraflow operations reduce the number of vehicles flowing on the normal-flow lanes and therefore reduce the merge congestion at downstream locations where vehicles from the on-ramp merge with the normal-flow lanes. This phenomenon was observed from the fact that a larger number of vehicles was unable to enter the network at Lakeland in Strategy 4 than in Strategy 3. The second aspect that should be kept in mind is the shockwave effect due to the slower speeds and the lower flow rates at the crossover. This case occurred at the crossover after Lakeland and could be attributed to the shockwave effect of vehicles at the crossover, because a large number of vehicles could not enter the network at Lakeland in Strategy 3, whereas all vehicles were able to enter the network at Lakeland in Strategy 1.

It was also found that CTM predicted the MOEs as well as VISSIM did, with an approximately 5% error in the results between CTM and VISSIM. With its faster computing time, CTM is ideal for use as a real-time decision support system and will be able to determine dynamic crossovers in the event of bottlenecks caused by incidents or vehicle breakdowns. The fast computation and realistic representation of traffic flow make it suitable for use for the modeling of evacuation routes and quick verification of the effectiveness of evacuation plans. CTM is also able to dissipate all vehicles into the network, allowing a more holistic comparison of the various strategies. With the need for more flexible evacuation plans, CTM's capability makes it an ideal tool for use in the testing of various scenarios.

TABLE 3 CTM Results for Each Strategy

CTM	Total Travel Time (h)	Evacuation Clearance Time (h)				Excess Vehicles at Lakeland
		t80	t85	t90	t95	
Base case	129,375	14.73	15.65	17.03	18.42	12,440
Strategy 1	109,583.3	13.91	15.13	16.68	18.24	0
Strategy 2	109,466.7	13.91	15.13	16.68	18.24	0
Strategy 3	114,191.7	14.32	15.48	16.91	18.35	8,386
Strategy 4	113,050	14.41	15.58	16.98	18.39	10,819

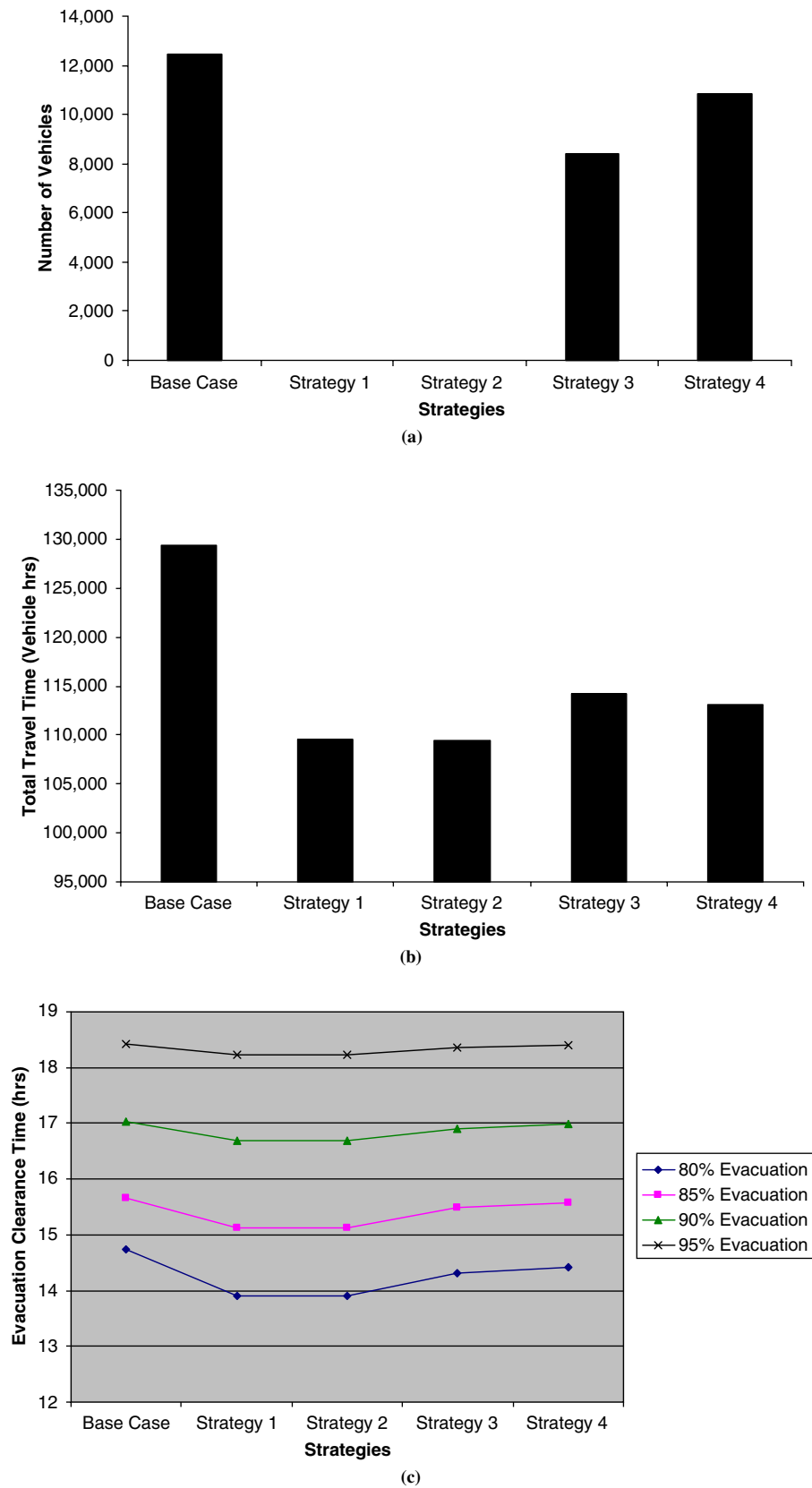


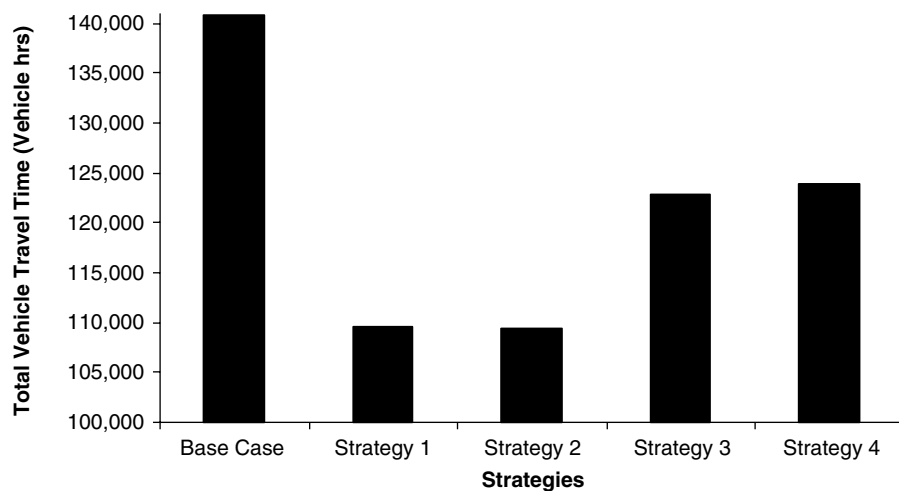
FIGURE 3 MOEs for various evacuation clearance time strategies obtained using CTM mesoscopic simulation: (a) number of vehicles not dissipated into the network for various strategies, (b) total vehicle travel time for various strategies, and (c) 80%, 85%, 90%, and 95% evacuation clearance times for various strategies.

TABLE 4 Percentage Errors Between Estimates of Microscopic Simulation (VISSIM) and Mesoscopic Simulation (CTM)

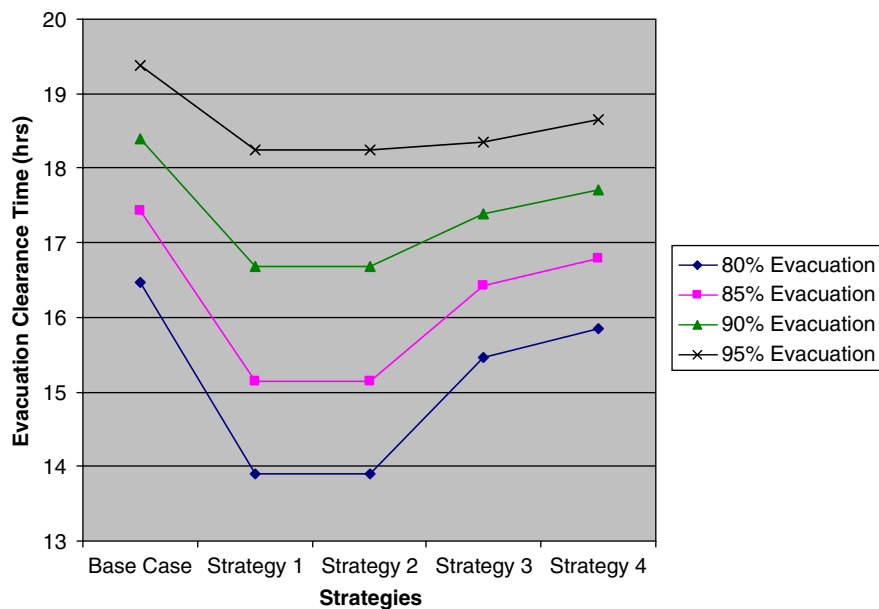
Errors	Total Travel Time (h)	t_{80}	t_{85}	t_{90}	t_{95}	Excess Vehicles at Lakeland
Base case	-3.46	-0.26	0.03	0.23	0.16	-4.41
Strategy 1	-3.18	1.05	1.49	0.93	0.47	0
Strategy 2	-2.31	1.27	1.81	0.96	0.3	0
Strategy 3	-4.37	0.99	1.27	0.78	0.48	-5.14
Strategy 4	-1.09	2.19	2.41	1.48	0.68	-2.83

TABLE 5 CTM Results for Each Strategy with Dissipation of All Vehicles into the Network

CTM	Total Travel Time (h)	Evacuation Clearance Time (h)				Excess Vehicles at Lakeland
		t_{80}	t_{85}	t_{90}	t_{95}	
Base case	140,825.00	16.46	17.43	18.4	19.38	0
Strategy 1	109,583.33	13.91	15.13	16.68	18.24	0
Strategy 2	109,466.67	13.91	15.13	16.68	18.24	0
Strategy 3	122,775.00	15.47	16.43	17.39	18.35	0
Strategy 4	123,816.67	15.85	16.78	17.72	18.65	0



(a)



(b)

FIGURE 4 MOEs for various evacuation clearance time strategies obtained using CTM mesoscopic simulation with all vehicles allowed to dissipate into the network: (a) total vehicle travel time for various strategies and (b) 80%, 85%, 90%, and 95% evacuation clearance time for various strategies.

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